

Prophet-Tune Project Specification

Overview

Create a complete Prophet-Tune time series forecasting web application from scratch.

Tech Stack

- **Frontend Framework:** React 19 with TypeScript
- **Build Tool:** Vite 7
- **Styling:** Tailwind CSS 3.4 with CSS variables for theming
- **UI Components:** shadcn-ui (built on Radix UI primitives)
- **Charts:** Recharts
- **Backend:** Supabase (auth, database, edge functions)
- **State Management:** TanStack React Query
- **Routing:** React Router DOM 7
- **Forms:** React Hook Form with Zod validation
- **Notifications:** Sonner toast library

Project Structure

```
prophet-tune/
├── package.json
├── vite.config.ts
├── tailwind.config.ts
├── tsconfig.json, tsconfig.app.json, tsconfig.node.json
├── postcss.config.js, eslint.config.js, index.html
├── src/
│   ├── main.tsx, App.tsx, index.css
│   ├── lib/utils.ts (cn helper)
│   ├── types/
│   │   ├── forecast.ts (ForecastModel, DataFrequency, PerformanceMetric)
│   │   ├── forecastResults.ts (ForecastPoint, MetricsResult)
│   │   ├── dataAnalysis.ts (stationarity, autocorrelation)
│   │   ├── hooks/useToast.ts, useMobile.ts
│   │   ├── utils/dataAnalysis.ts (MAE, RMSE, MAPE, R-squared)
│   │   ├── integrations/supabase/client.ts, types.ts
│   │   ├── pages/Index.tsx, Auth.tsx, NotFound.tsx
│   │   └── components/
│   │       ├── ui/ (18 shadcn-style components)
│   │       └── forecast/ (8 workflow components)
```

UI Components (src/components/ui/)

button.tsx, label.tsx, input.tsx, card.tsx, tabs.tsx, select.tsx, checkbox.tsx, radio-group.tsx, switch.tsx, slider.tsx, progress.tsx, table.tsx, badge.tsx, tooltip.tsx, separator.tsx, scroll-area.tsx, alert.tsx, toast.tsx

Forecast Components (src/components/forecast/)

- **DataUpload.tsx:** CSV upload with drag-drop, PapaParse parsing
- **ModelSelector.tsx:** Prophet, AutoGluon, ARIMA, AR, ARMA selection
- **VariableConfig.tsx:** Date, segment, dependent variable selection

- **SegmentMapper.tsx**: Per-segment train/test/forecast configuration
- **ProphetHyperparameters.tsx**: Growth, changepoints, seasonality, uncertainty
- **MetricsSelector.tsx**: MAE, RMSE, MAPE, SMAPE, MSE, R-squared, coverage, MASE
- **ForecastProgress.tsx**: Progress indicator during forecast
- **ForecastResults.tsx**: Charts with Recharts, metrics table, export

9-Step Workflow

1. **Upload Data** - CSV file upload with preview
2. **Select Model** - Choose forecasting model (Prophet default)
3. **Configure Variables** - Set date column, segment column, dependent variable
4. **Segment Mapping** - Configure train/test split, forecast periods per segment
5. **Data Analysis** - AI-powered transformation recommendations (placeholder)
6. **Regressors** - External regressor configuration (placeholder)
7. **Metrics** - Select performance metrics to calculate
8. **Parameters** - Model-specific hyperparameters (Prophet: growth, seasonality)
9. **Results** - Interactive charts, metrics table, export options

Key Features

- Drag-and-drop CSV upload with data preview
- Multi-segment forecasting support
- Prophet hyperparameter configuration UI with sliders
- Interactive forecast charts with confidence intervals
- Train/test split visualization
- Performance metrics comparison
- Dark/light theme support via CSS variables
- Mobile-responsive design

Key Dependencies (package.json)

- @radix-ui/* (accordion, checkbox, dialog, dropdown-menu, label, popover, progress, radio-group, scroll-area, select, separator, slider, slot, switch, tabs, toast, toggle, tooltip)
- @supabase/supabase-js, @tanstack/react-query
- class-variance-authority, clsx, tailwind-merge
- date-fns, lucide-react (icons), papaparse (CSV)
- react, react-dom, react-router-dom
- react-hook-form, @hookform/resolvers, zod
- recharts, sonner, tailwindcss-animate

Instructions for New Chat

When asked to create this project, generate a complete, working React application with all the files listed above. Use the shadcn-ui component patterns (Radix primitives with CVA styling). Include mock forecast functionality that generates sample data for demonstration.